



Off-Grid PV Battery Charger

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Abstract

Remote off-grid systems often rely on batteries to power mechanical loads. This project proposes a solar-powered system for recharging a 12V lead-acid battery while simultaneously powering a connected mechanical load. The solar panel generates the power required for both battery charging and mechanical load operation, and the charged battery supplies additional power when solar generation alone is insufficient. To maximize power generation, a maximum power point tracking (MPPT) algorithm using the perturb and observe (P&O) method is used. The MPPT algorithm is implemented on a microcontroller and regulates the panel voltage through a buck converter. To maintain safe charging and prolong battery lifespan, a 4-step lead-acid battery charging algorithm is implemented in a secondary buck-boost charge controller.

System Overview

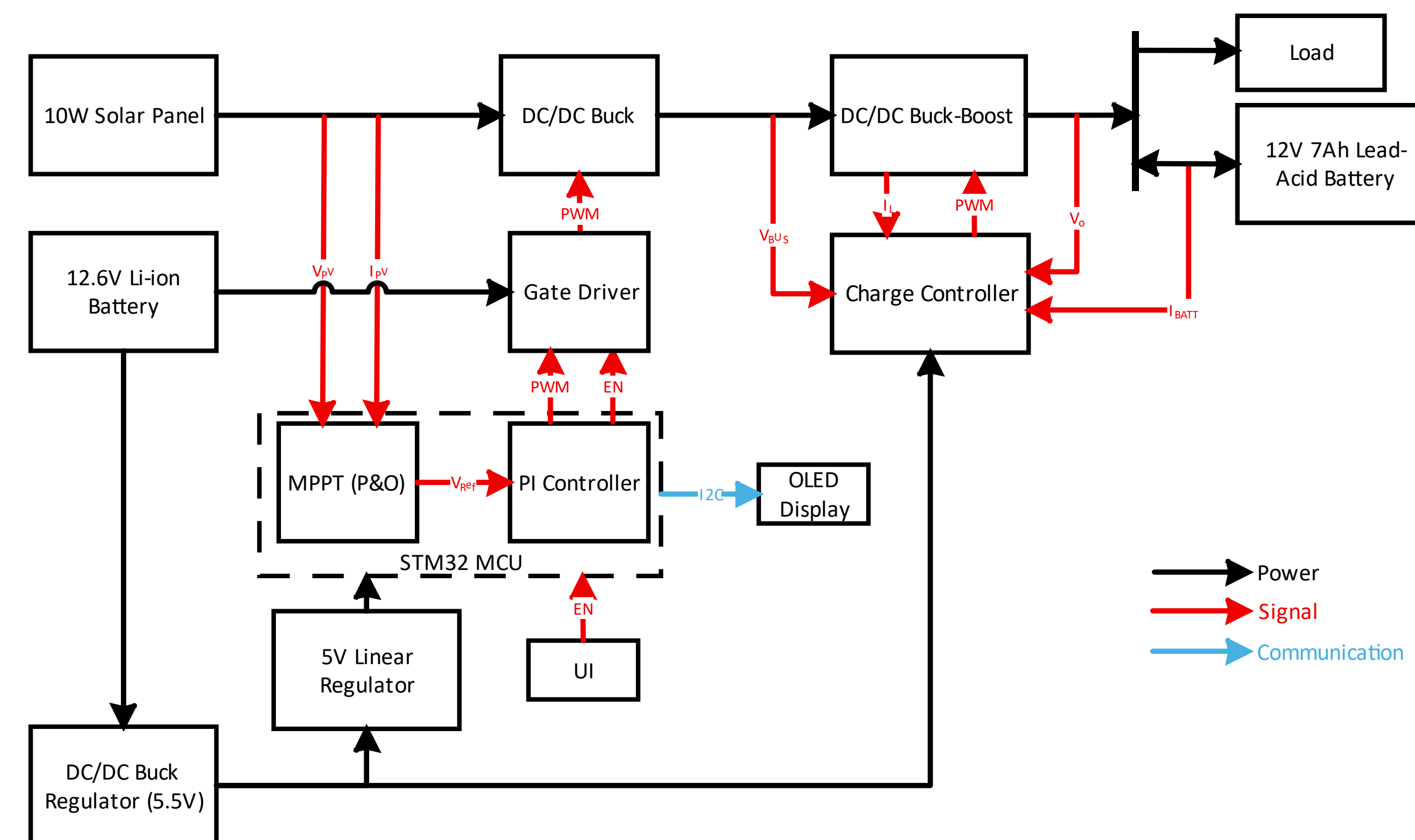


Figure 1: System block diagram of battery charger

10 W solar panel regulated to V_{mp} using a synchronous buck converter. A 4-switch buck-boost converter performs precondition, bulk, absorption, and float charging states for a 12 V lead-acid battery while supplying a connected load.

Features:

- Sensing:** Voltage is monitored with resistor dividers, and current is measured with INA296A5 current sensors. Sensor outputs read by STM32G4 ADC channels.
- Regulation:** P&O MPPT regulates V_{PV} to V_{mp} . V_{out} regulated to 13.3 V when battery is absent or deeply discharged.
- Limits/Protection:** I_{BATT} limited to 1.4 A. Intermediate bus undervoltage cutoff at 5 V. Timed battery preconditioning with fault if voltage recovery does not occur.
- Control:** STM32G4 implements MPPT and PI control. LTC4020 manages battery charging
- Indicators:** LEDs indicate the active charging state, and an OLED display provides real time voltage and current measurements.

Results

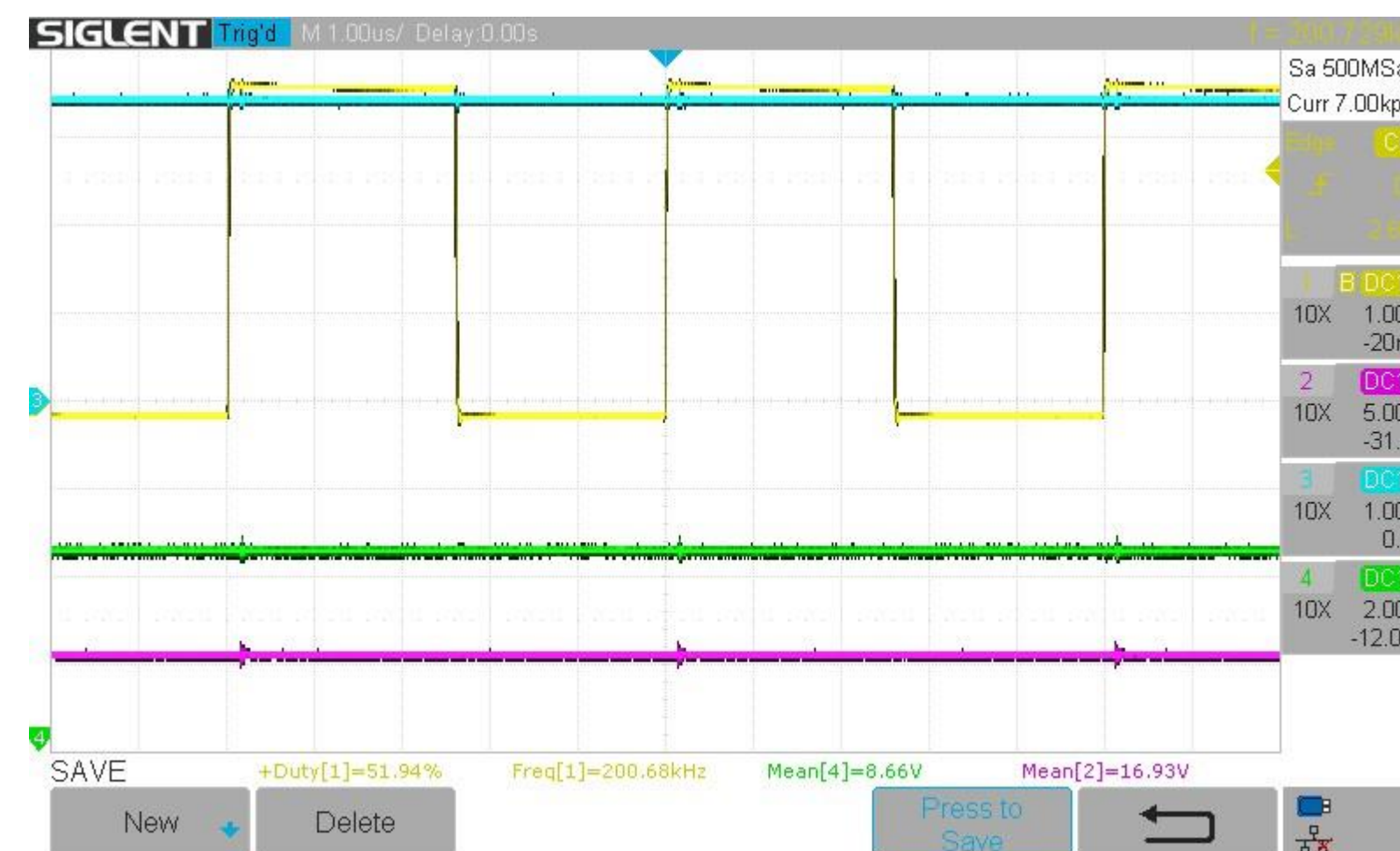


Figure 2: V_{PV} (pink), V_{BUS} (green), enable signal (blue), and MPPT Buck PWM (yellow) waveforms during closed-loop MPPT operation

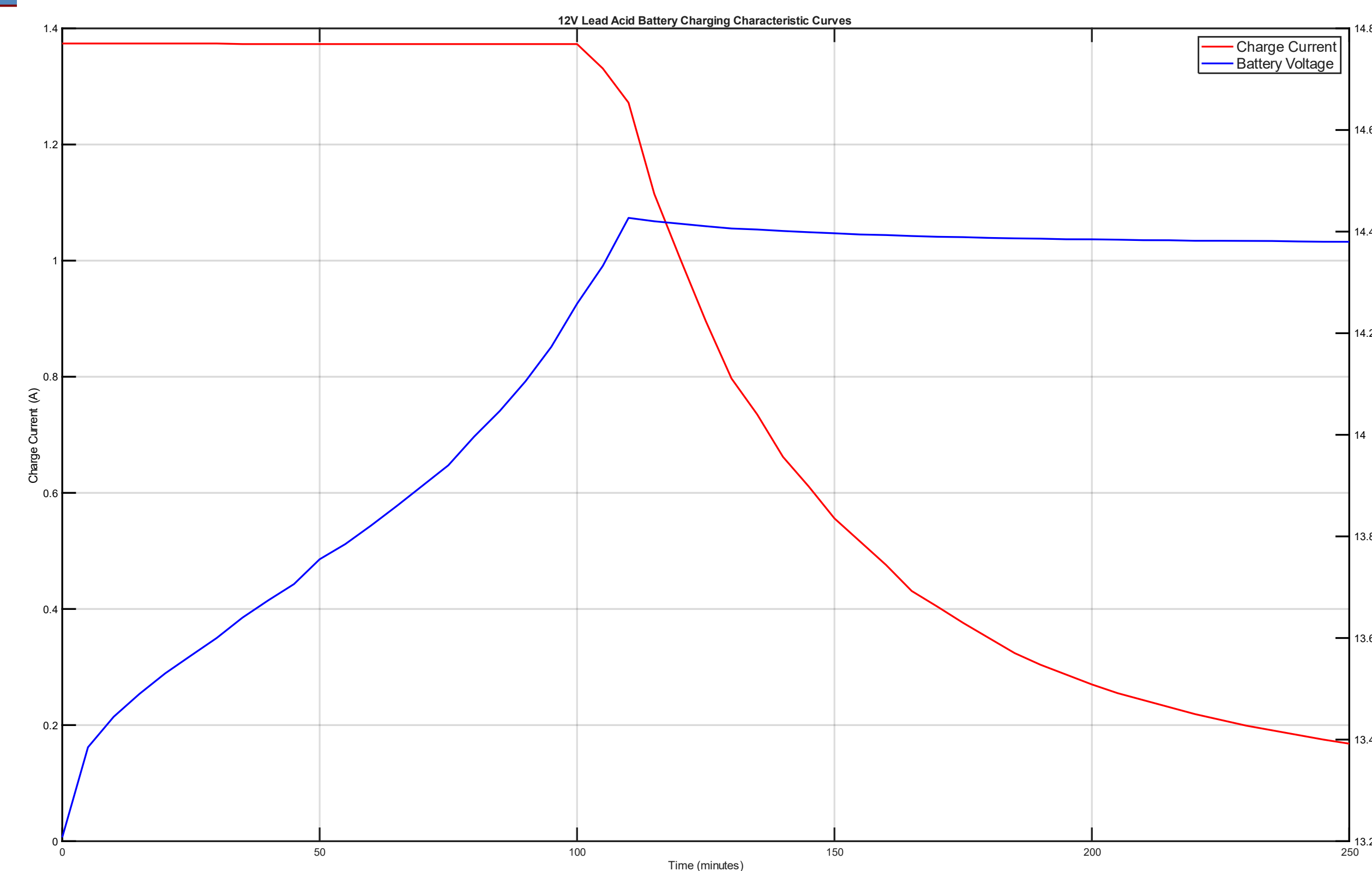


Figure 5: Measured I_{BATT} and V_{BATT} characteristics during charging of 12V 7Ah lead-acid battery using the developed charge controller PCB

- Hardware:
 - MPPT buck PCB (4 layers, 64 x 39 mm)
 - Charge controller PCB (6 layers, 80 x 93.4 mm)

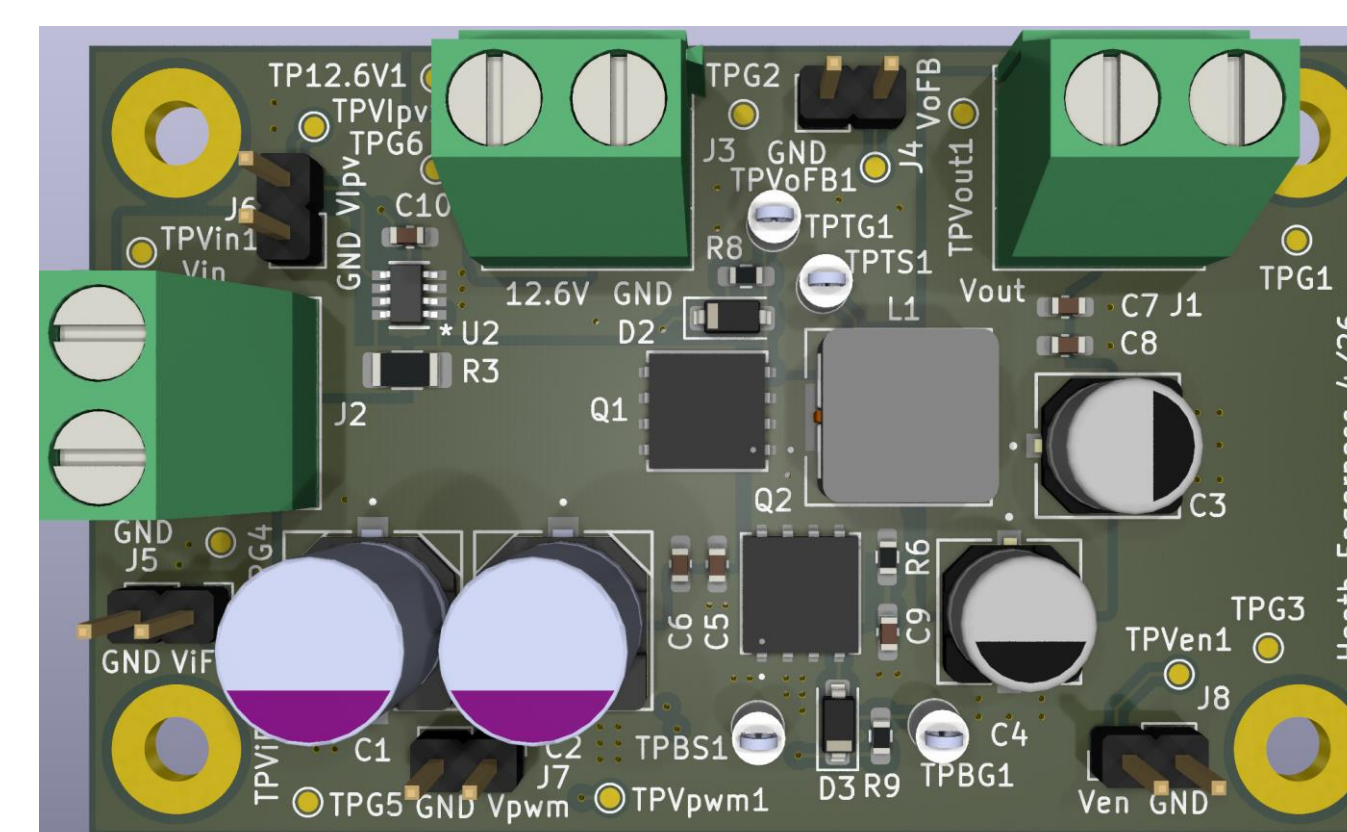


Figure 6: MPPT controller PCB

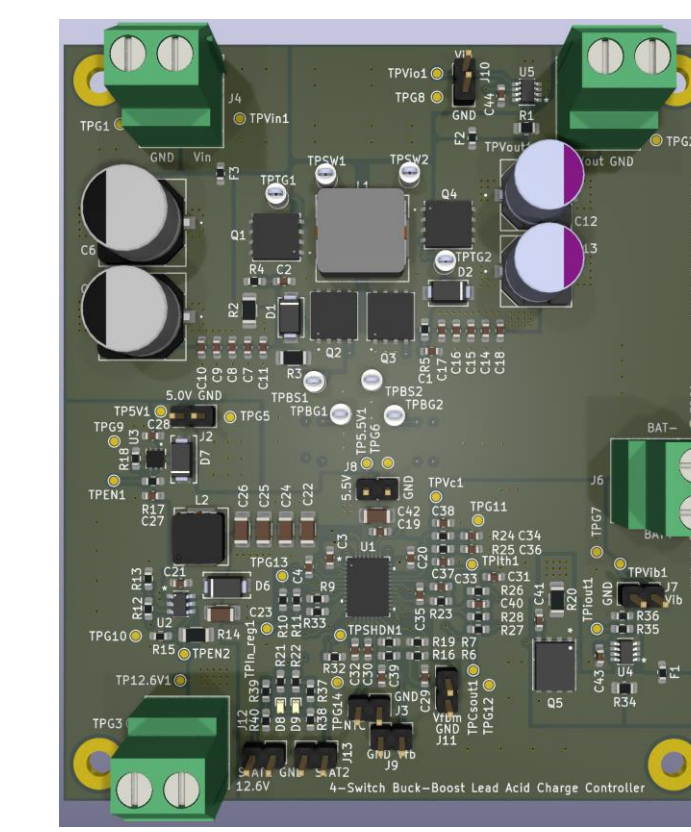


Figure 7: Charge controller PCB

- MPPT algorithm tracks and regulates V_{PV} to V_{mp} , which is 17.5 V under standard test conditions (STC)
- V_{BUS} remains stable during load operation
- Sensed V_{PV} , V_{BUS} , and I_{PV} , and calculated P_{PV} are monitored on OLED display
- Push button enable (EN) allows user controlled start up and shut down of the MPPT buck
- MPPT buck converter efficiency: 93.49%

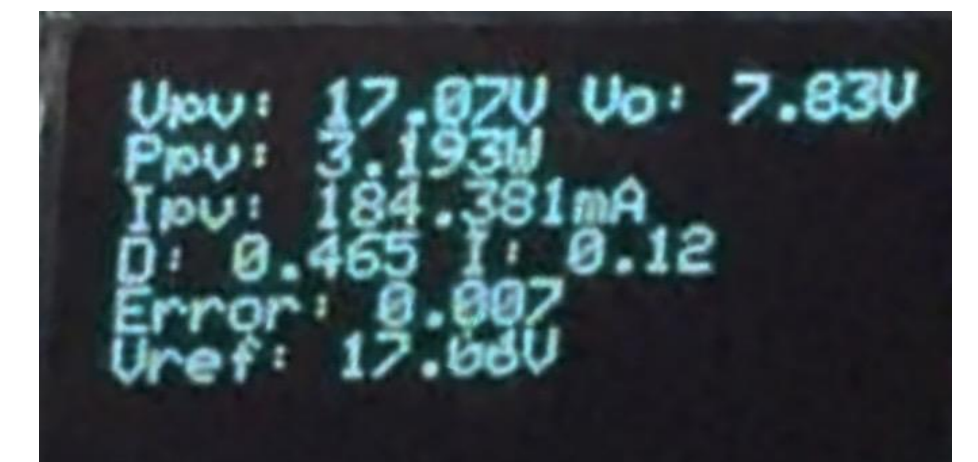
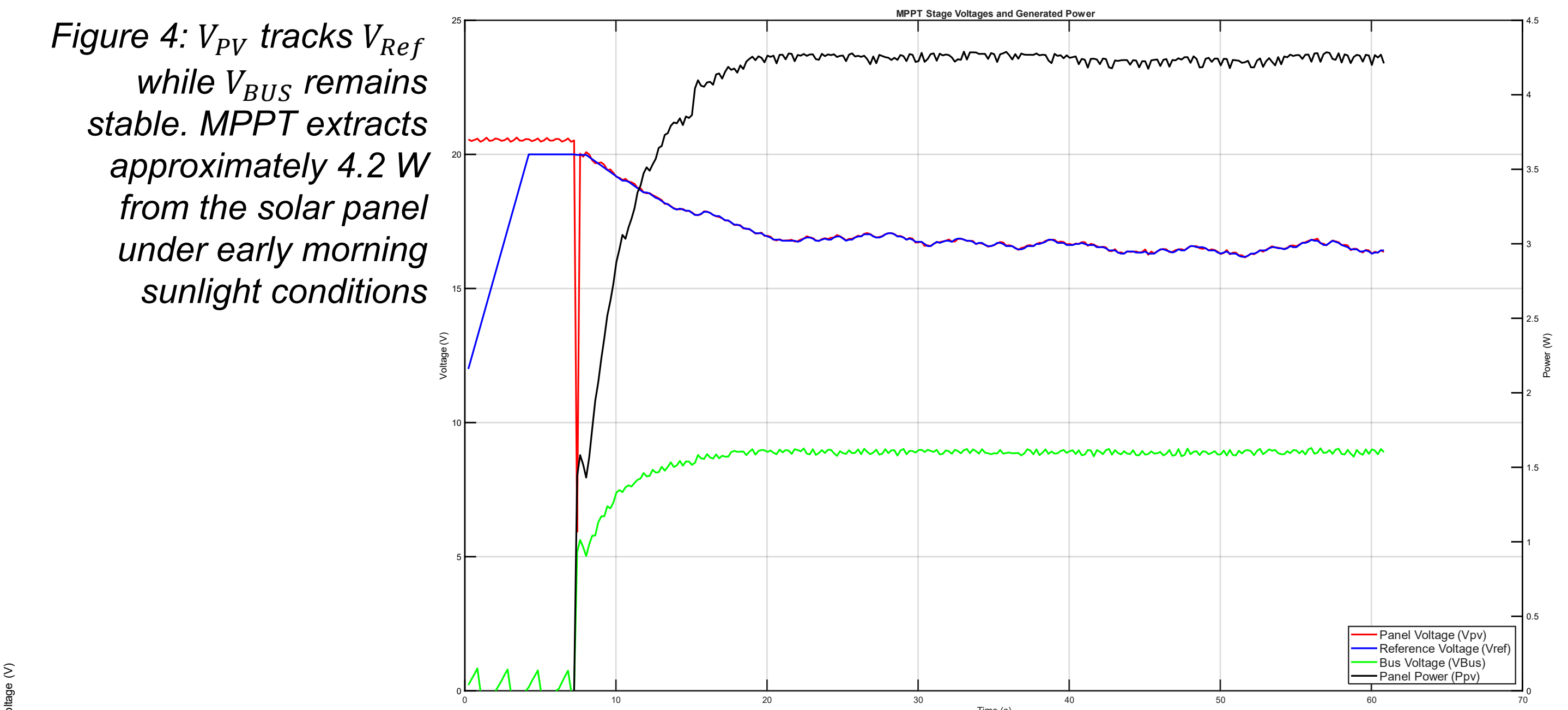


Figure 3: OLED display demonstrating MPPT operation and regulation of V_{PV} to V_{MP} under cloudy conditions



- Initial battery voltage: 12.35V ($\approx 50\%$ SOC)
- Bulk:** $I_{BATT} = 1.374 A$ until $V_{BATT} = 14.41 V$, then charge transition to absorption
- Absorption:** I_{BATT} steadily decreased as battery approached full charge, then charge transition to float
- Float:** $I_{BATT} < 140 mA$ (10% of charge current limit)
 - Observed float transition current: 168 mA
- Status indicators transition for each charge state
- Total charging time until float termination: 4 hours 10 minutes

- V_{out} regulated to 13.3 V when lead-acid battery is not present
- Lead-acid battery supplied additional power to heavy load when solar power insufficient
- Buck-boost converter efficiency by operating mode:
 - Buck: 90.82%
 - Boost: 90.16%
 - Buck-Boost: 90.7%

Future Direction

- Expand charging support for additional battery chemistries, like Li-ion and LiFePO4
- Add battery temperature monitor for increased charging safety
- Add support for USB charging ports for powering or charging portable electronics